AERSP 301 — closed sections: finding $q_{s,0}$

Setup (always)

Cut the closed section at a convenient point. The shear flow is $q_s = q_b + q_{s,0}$, where q_b is the open-section "basic" flow (zero at the cut) and $q_{s,0}$ is the unknown constant that closes the loop. Every method below only finds $q_{s,0}$; they differ in *which condition* fixes it.

1. General moment equation (load anywhere, position known)

$$S_x\,\eta_0 - S_y\,\xi_0 = \oint p\,q_b\,ds + 2A\,q_{s,0} \implies q_{s,0} = \frac{S_x\eta_0 - S_y\xi_0 - \oint p\,q_b\,ds}{2A}$$

p is the perpendicular distance from the moment center to the tangent of the wall; A is the area enclosed by the wall midline; (ξ_0, η_0) locate the line of action of the load relative to the moment center.
Use when the load acts at a *known* position, and you do not want to pick a clever moment center.

2. Moment center on the load's line of action

$$q_{s,0} = -\frac{\oint p\,q_b\,ds}{2A}$$

Special case of (1): choose the moment center so the applied moment is zero ($\xi_0 = \eta_0 = 0$).
Use when the load passes through a point you can name, e.g. a vertex, an apex, or the shear center. *This is the one for HW 3, Problem 2:* the load is applied at the apex, so take moments about the apex. Walls through the apex have $p = 0$ and drop out.

3. Zero twist (load through the shear center)

$$\frac{d\theta}{dz} = \frac{1}{2A}\oint \frac{q_s}{Gt}\,ds = 0 \implies q_{s,0} = -\frac{\oint (q_b/Gt)\,ds}{\oint ds/(Gt)}$$

Use when the load acts through the shear center (no twist), or you are *solving for* the shear center. Valid for any G and t that vary around the section. No moment center needed.

4. Zero twist, Gt constant

$$q_{s,0} = -\frac{\oint q_b\,ds}{\oint ds}$$

Special case of (3). **Use when** G and t are the same on every wall. Per-wall thickness differences (e.g. one wall $2t$) mean you must use (3), not (4).

How they fit together: shear center of a closed section

1. Use (3) or (4) to get $q_{s,0}$, hence $q_s = q_b + q_{s,0}$ (load through the shear center means no twist).
2. Take moments of this q_s about any convenient point to get the shear-center coordinate: $S_y\,\xi_s = \oint p\,q_s\,ds$ (or the η_s version for S_x).
3. Check: methods (2) and (3)/(4) give the *same* $q_{s,0}$ only if the load really is at the shear center. A load elsewhere twists the section, and then (2) is the right one.